Fiscal Year:	FY 2015	**Task Last Updated:**	FY 12/12/2014
PI Name:	Kuczmarski, James M. Ph.D.		
Project Title:	Developing Novel, Targeted Countermeasures to Reduce Oxidative Stress and Skeletal Muscle Atrophy During Microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Musculoskeletal Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-4243	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 NSBRI-RFA-13-01 Postdoctoral Fellowships
Start Date:	12/01/2013	**End Date:**	11/30/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lawler, John Ph.D. (MENTOR/ Texas A&M University)		
Grant/Contract No.:	NCC 9-58-PF03504		
Performance Goal No.:			
Performance Goal Text:			
	POSTDOCTORAL FELLOWSHIP The reduced mechanical loading of spaceflight is associated with decreased skeletal muscle fiber cross sectional area (CSA) and force-generating capacity. Muscle atrophy and weakness may then impede mission objectives, increase the risk of injury, and increase susceptibility to damage upon reloading. Muscular alterations with spaceflight are therefore a critical limitation to performance during spaceflight missions and a serious health concern for astronauts. Recent evidence suggests that oxidative stress is increased in microgravity and contributes to skeletal muscle atrophy. Oxidative stress is classically defined as an imbalance between reactive oxygen species (ROS) production and removal by endogenous antioxidants. ROS are produced primarily from the mitochondria as well as nonphagocytic NAPDH-oxidases (Nox) in skeletal muscle and can have various harmful effects without adequate antioxidant		

Task Description:	sequestration. ROS have been shown to activate different proteolytic pathways during skeletal muscle disuse and may induce translocation of the muscle specific neuronal nitric oxide synthase-μ (nNOSμ) splice variant from the sarcolemma to the cytosol leading to muscle atrophy. The purpose of this investigation is to develop novel, targeted countermeasures to abolish the effects of oxidative stress and prevent muscle atrophy during microgravity. Our experimental design consists of a series of studies utilizing a 7-day hindlimb-unloading (HU) rat model of microgravity to evaluate the therapeutic effectiveness of different antioxidants/ROS-inhibitors. In the original research proposal, two specific aims were listed: (1) determine the efficacy of countermeasures that raise HSP70 against unloading-induced muscle atrophy; (2) identify the contribution of renin-angiotensin II signaling to oxidative stress and muscle atrophy with mechanical unloading by utilizing the angiotensin-II receptor blocker, losartan. Overall Project's Objectives: Dr. John Lawler will serve as a mentor for James Kuczmarski in a rich research environment at Texas A&M, including the Space Life Science Program. Dr. Kuczmarski will learn the latest molecular and image analysis tools in the development of highly novel countermeasures against skeletal muscle atrophy during microgravity. Dr. Lawler's laboratory is supported by the NASA Space Biology program, and is dedicated to finding targeted, antioxidant countermeasures against spaceflight sarcopenia. Key Findings: We have begun to test the therapeutic effectiveness of losartan as well as: a potent antioxidant, Eukarion-134 (EUK-134), a Nox peptidyl inhibitor, gp91ds-tat, and the mitochondrial ROS peptidyl inhibitor, SS-31. Data that has been collected thus far indicates that: EUK-134 decreased muscle ROS production as indicated by lower dihydroethidium oxidation, abolished skeletal muscle atrophy, and prevented the shift in fiber type from slow to fast. EUK-134 also increased expression of nNOS at the membrane. Anabolic signaling via the protein kinase, Akt, was improved with EUK-134. Similar to EUK-134, gp91 ds-tat and SS-31 improved soleus muscle mass and fiber CSA. Losartan did not improve soleus muscle mass but increased Type I fiber CSA. Conclusions and Future Directions: Inhibition of ROS is effective in limiting skeletal muscle atrophy with mechanical unloading; therefore, antioxidants/ROS-inhibitors may represent translatable treatment options to prevent skeletal muscle atrophy in astronauts experienced with spaceflight. Future study will focus on finalizing gp91 ds-tat, SS-31, and Losartan interventional studies as well as establishing the effects of HSP70 agonists on oxidative stress in skeletal muscle with mechanical unloading. Dr. Kuczmarski recently accepted a postdoctoral fellowship position at the Penn State Hershey Medical Center. His future work will primarily involve investigation of skeletal muscle blood flow with the hope to eventually extrapolate study to different models of microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overall goal of this project is to eventually translate our findings to astronauts and implement these pharmaceutical interventions during spaceflight with the intention of improving performance through increased skeletal muscle mass and related function as outlined in the NASA Human Research Program Evidence Reports. In addition, similarities may exist between the mechanisms underlying skeletal muscle atrophy experienced in microgravity and related muscular disorders on Earth including aging-related sarcopenia and muscular dystrophy. Therefore, it is plausible that developed antioxidant countermeasures may also benefit those suffering on Earth.
Task Progress:	We have made significant progress on the project's overall goal in developing novel, targeted countermeasures to reduce oxidative stress and skeletal muscle atrophy during microgravity. This includes progress on the following subprojects: (1) Administration of EUK-134 prevents 7-day unloading induced atrophy. A paper is in preparation demonstrating that administration of the cell permeable superoxide dismutase/catalase mimetic, Eukarion-134, decreases muscle ROS production as indicated by lower dihydroethidium oxidation. In doing so, EUK-134 abolished skeletal muscle atrophy as measured by muscle mass and Type I fiber cross sectional area. Inhibition of muscle atrophy was related to increased expression of nNOS at the membrane in conjunction with associated proteins, dysferlin and caveolin-3. In addition, EUK-134 increased anabolic signaling via the serine/threonine-specific protein kinase, Akt. Specifically, Akt serine 473 and mTOR serine 2448 phosphorylation were increased in HU-EUK-134 animals and p70-S6k threonine 389 phosphorylation was preserved. (2) More targeted peptidyl inhibitors of ROS also prevent muscle atrophy with mechanical unloading and suggest the source of ROS production. The peptidyl NADPH-oxidase-2 (Nox2) inhibitor, gp91 ds-tat, and a peptidyl mitochondrial ROS inhibitor, SS-31, are being utilized during 7D hindlimb unloading protocols, and data collected thus far indicates that both inhibitors improve soleus muscle mass and fiber cross sectional area. This indicates that the source of ROS may be both Nox2 and mitochondria during simulated microgravity. (3) The angiotensin II receptor blocker, Losartan, prevents the reduction in soleus muscle fiber cross sectional area following 7D of hindlimb unloading. Angiotensin II has been identified as a potent stimulus for ROS production in skeletal muscle via the G-protein coupled angiotensin type 1 receptor (AT1R). The AT1 receptor blocker, Losartan is therefore also being utilized for molecular targeting of ROS. Data collected suggests that Losartan is effective in preventing the reduction in soleus muscle fiber cross sectional area.
Bibliography Type:	Description: (Last Updated: 05/27/2020)
Abstracts for Journals and Proceedings	Lawler JM, Lee Y, Hord JM, Guzzoni V, Kuczmarski JM. "Mitochondrial ROS Amplify Nox2 Signaling and Atrophy in the Rat Soleus with 7 Days of Hindlimb Unloading." Experimental Biology 2014, San Diego, CA, April 26-30, 2014. FASEB Journal. 2014 Apr;28(1 Suppl):LB821. See also http://www.fasebj.org/content/28/1_Supplement.toc to search and locate abstract. , Apr-2014
Articles in Peer-reviewed Journals	Kuczmarski JM, Hord JM, Lee Y, Guzzoni V, Rodriguez D, Lawler MS, Garcia-Villatoro EL, Holly D, Ryan P, Falcon K, Garcia M, Janini Gomes M, Fluckey JD, Lawler JM. "Effect of Eukarion-134 on Akt-mTOR signaling in the rat soleus during 7 days of mechanical unloading." Exp Physiol. 2018 Apr 1;103(4):545-58. Epub 2018 Feb 28. https://doi.org/10.1113/EP086649 ; PubMed PMID: 29315934 , Apr-2018

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